1 Machine Learning

1.1 Supervised Learning

1. Regression Problems
2. Classification Problems
3. Simple Linear Regression
4. Multiple Linear Regression
5. Ridge Regression
6. Logistic Regression
7. K-Nearest Neighbour (KNN)
8. Naive Bayes Classifier
9. Linear Discriminant Analysis (LDA)
10. Support Vector Machine (SVM)
11. Decision Trees
12. Bias-Variance Trade-off
13. Leave-One-Out (LOO) Cross Validation
14. K-Fold Cross Validation
15. Multi-Layer Perceptron (MLP)
16. Feed-Forward Neural Network (FFNN)

1.2 Unsupervised Learning

1. Clustering Algorithms
2. K-Means Clustering
3. K-Medoids Clustering
4. Hierarchical Clustering
5. Top-Down Clustering
6. Bottom-Up Clustering
7. Single Linkage
8. Multiple Linkage
9. Dimensionality Reduction
10. Principal Component Analysis (PCA)